
Data Analysis

What is Data?

Collection of data objects and their attributes

An attribute is a property or characteristic of an object

Examples: eye color of a person, temperature, etc.

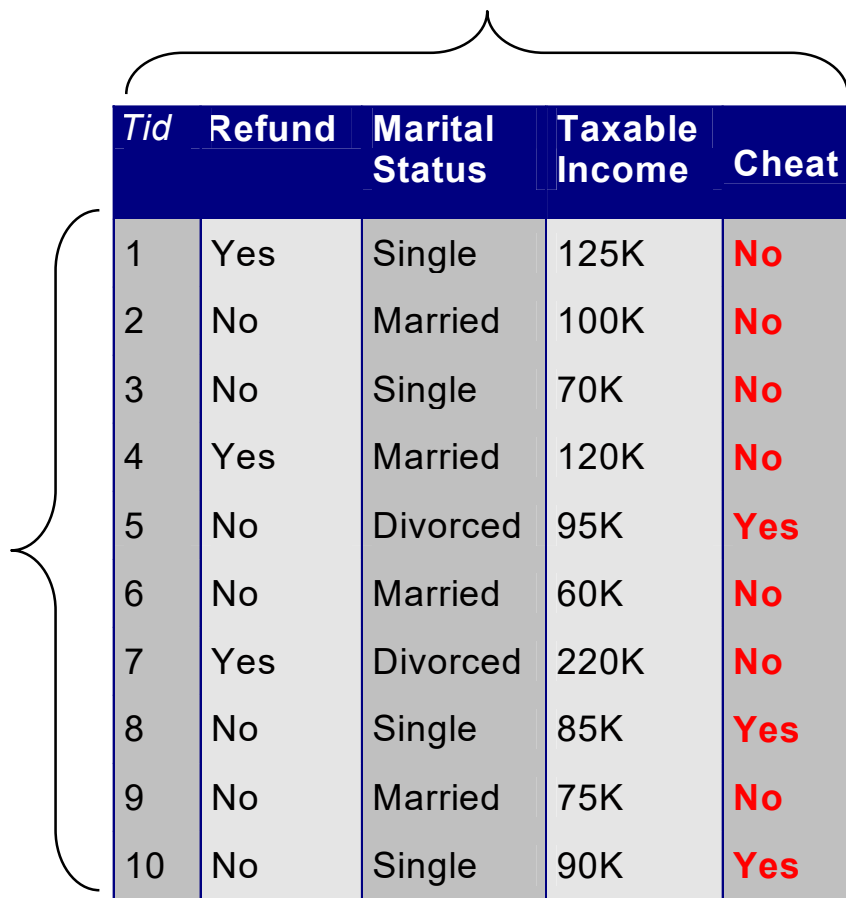
Attribute is also known as variable, field, characteristic, or feature

A collection of attributes describe an object

Object is also known as record, point, case, sample, entity, or instance

Attributes

Objects



<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attribute Values

Attribute values are numbers or symbols assigned to an attribute

Distinction between attributes and attribute values

Same attribute can be mapped to different attribute values

Example: height can be measured in feet or meters

Different attributes can be mapped to the same set of values

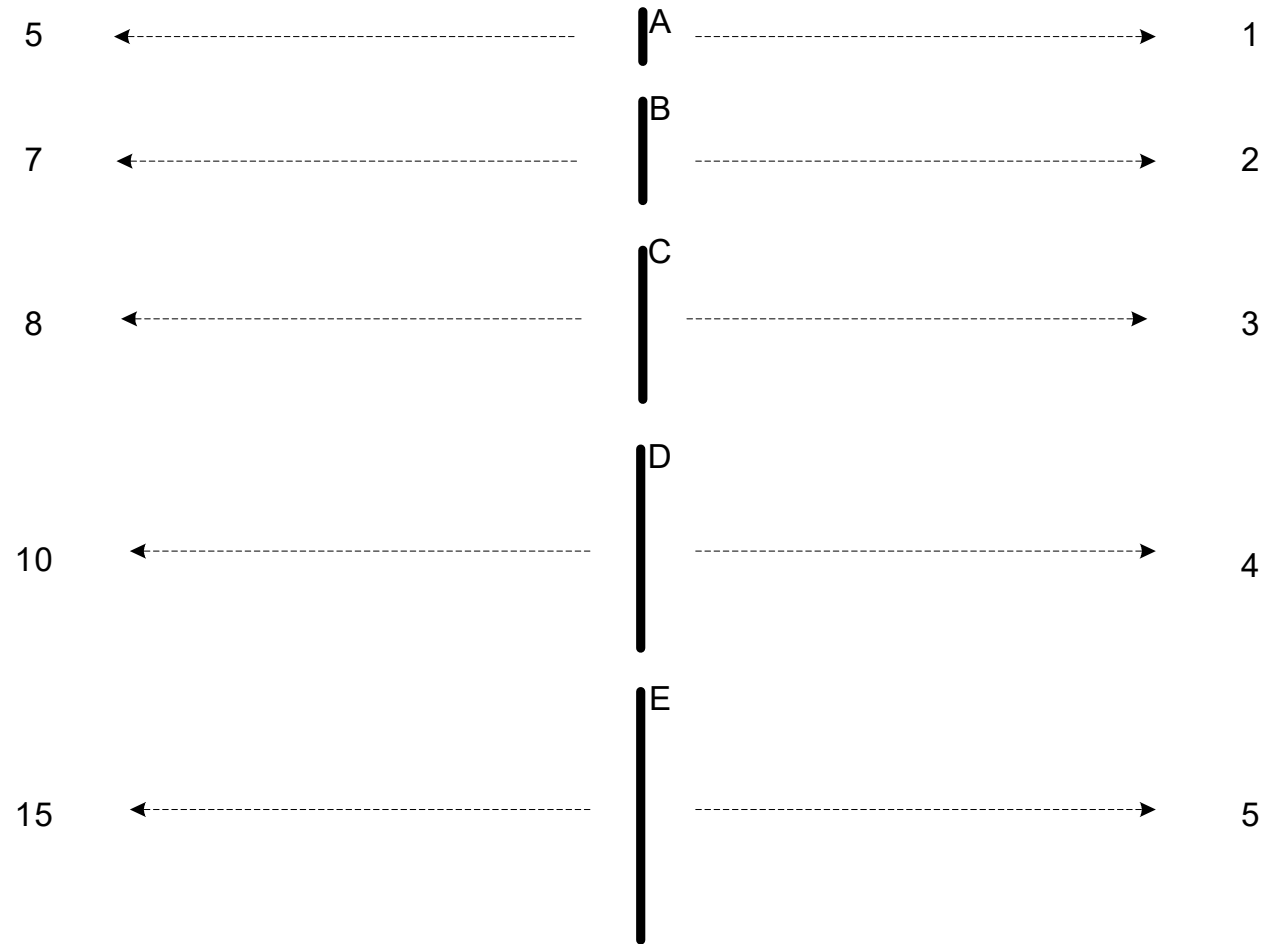
Example: Attribute values for ID and age are integers

But properties of attribute values can be different

ID has no limit but age has a maximum and minimum value

Measurement of Length

The way you measure an attribute is somewhat may not match the attributes properties.



Types of Attributes

There are different types of attributes

Nominal

Examples: ID numbers, eye color, zip codes

Ordinal

Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}

Interval

Examples: calendar dates, temperatures in Celsius or Fahrenheit.

Ratio

Examples: temperature in Kelvin, length, time, counts

Properties of Attribute Values

The type of an attribute depends on which of the following properties it possesses:

Distinctness: $= \neq$

Order: $< >$

Addition: $+ -$

Multiplication: $* /$

Nominal attribute: distinctness

Ordinal attribute: distinctness & order

Interval attribute: distinctness, order & addition

Ratio attribute: all 4 properties

Attribute Type	Description	Examples	Operations
Nominal	The values of a nominal attribute are just different names, i.e., nominal attributes provide only enough information to distinguish one object from another. ($=$, $\neq$)	zip codes, employee ID numbers, eye color, sex: $\{male, female\}$	mode, entropy, contingency correlation, χ^2 test
Ordinal	The values of an ordinal attribute provide enough information to order objects. ($<$, $>$)	hardness of minerals, $\{good, better, best\}$, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Interval	For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists. ($+$, $-$)	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, t and F tests
Ratio	For ratio variables, both differences and ratios are meaningful. ($*$, $/$)	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current	geometric mean, harmonic mean, percent variation

Attribute Level	Transformation	Comments
Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
Ordinal	An order preserving change of values, i.e., $new_value = f(old_value)$ where f is a monotonic function.	An attribute encompassing the notion of good, better best can be represented equally well by the values {1, 2, 3} or by { 0.5, 1, 10}.
Interval	$new_value = a * old_value + b$ where a and b are constants	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
Ratio	$new_value = a * old_value$	Length can be measured in meters or feet.

Discrete and Continuous Attributes

Discrete Attribute

Has only a finite or countably infinite set of values

Examples: zip codes, counts, or the set of words in a collection of documents

Often represented as integer variables.

Note: binary attributes are a special case of discrete attributes

Continuous Attribute

Has real numbers as attribute values

Examples: temperature, height, or weight.

Practically, real values can only be measured and represented using a finite number of digits.

Continuous attributes are typically represented as floating-point variables.

Types of data sets

Record

Data Matrix

Document Data

Transaction Data

Graph

World Wide Web

Molecular Structures

Ordered

Spatial Data

Temporal Data

Sequential Data

Genetic Sequence Data

Record Data

Data that consists of a collection of records, each of which consists of a fixed set of attributes

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
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3	No	Single	70K	No
4	Yes	Married	120K	No
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7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Data Matrix

If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute

Such data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Document Data

Each document becomes a 'term' vector,

- each term is a component (attribute) of the vector,
- the value of each component is the number of times the corresponding term occurs in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Transaction Data

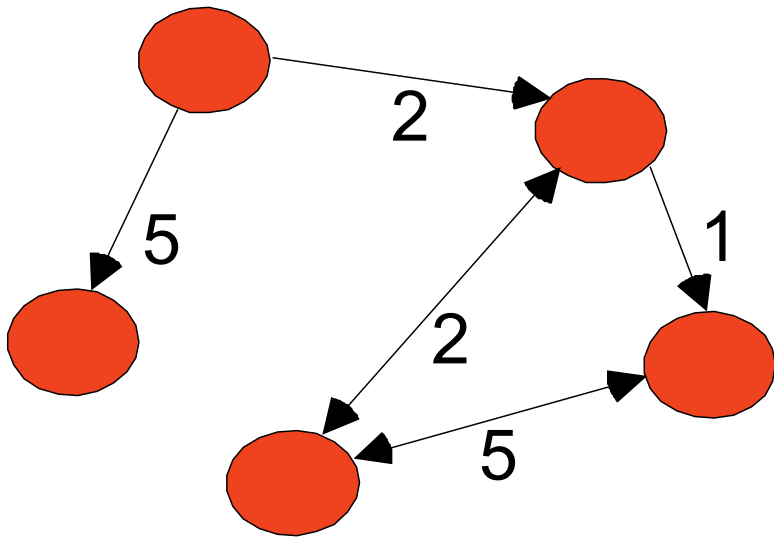
A special type of record data, where each record (transaction) involves a set of items.

For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Graph Data

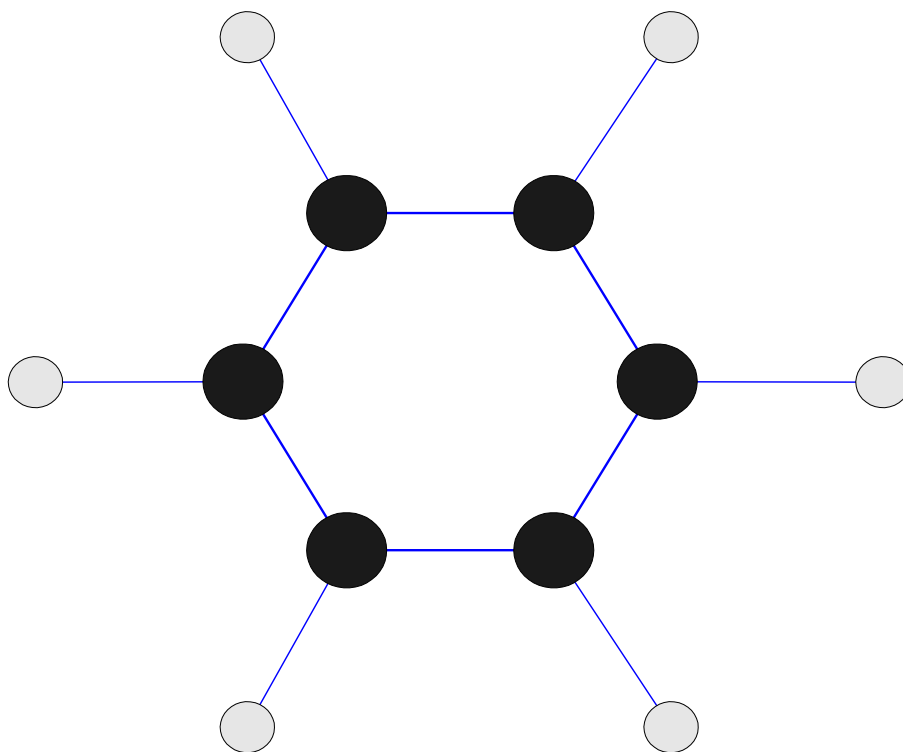
Examples: Generic graph and HTML Links



```
<a href="papers/papers.html#bbbb">  
Data Mining </a>  
<li>  
<a href="papers/papers.html#aaaa">  
Graph Partitioning </a>  
<li>  
<a href="papers/papers.html#aaaa">  
Parallel Solution of Sparse Linear System of Equations </a>  
<li>  
<a href="papers/papers.html#ffff">  
N-Body Computation and Dense Linear System Solvers
```

Chemical Data

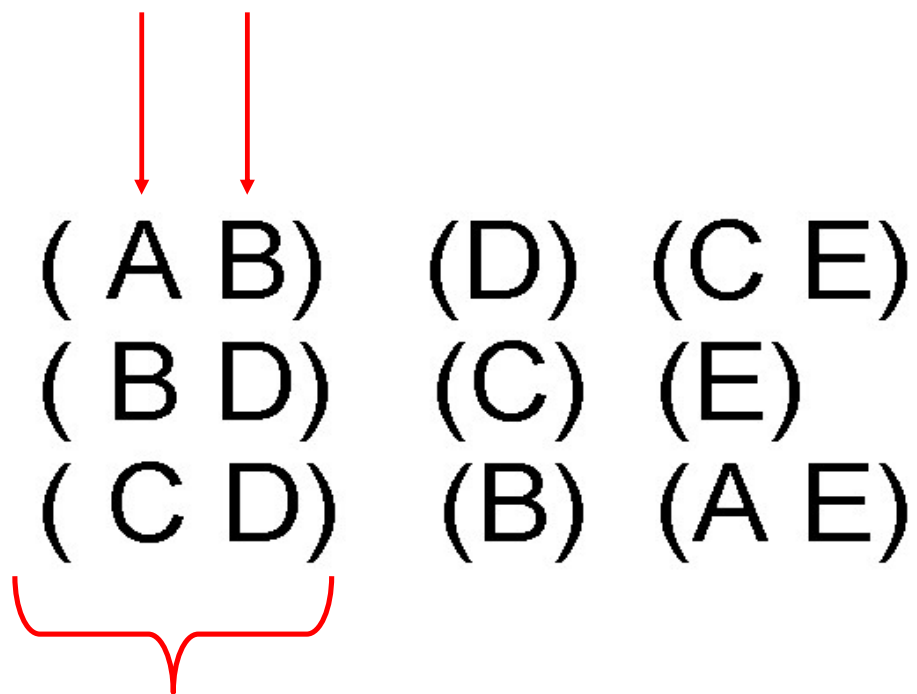
Benzene Molecule: C_6H_6



Ordered Data

Sequences of transactions

Items/Events



An element of
the sequence

Ordered Data

Genomic sequence data

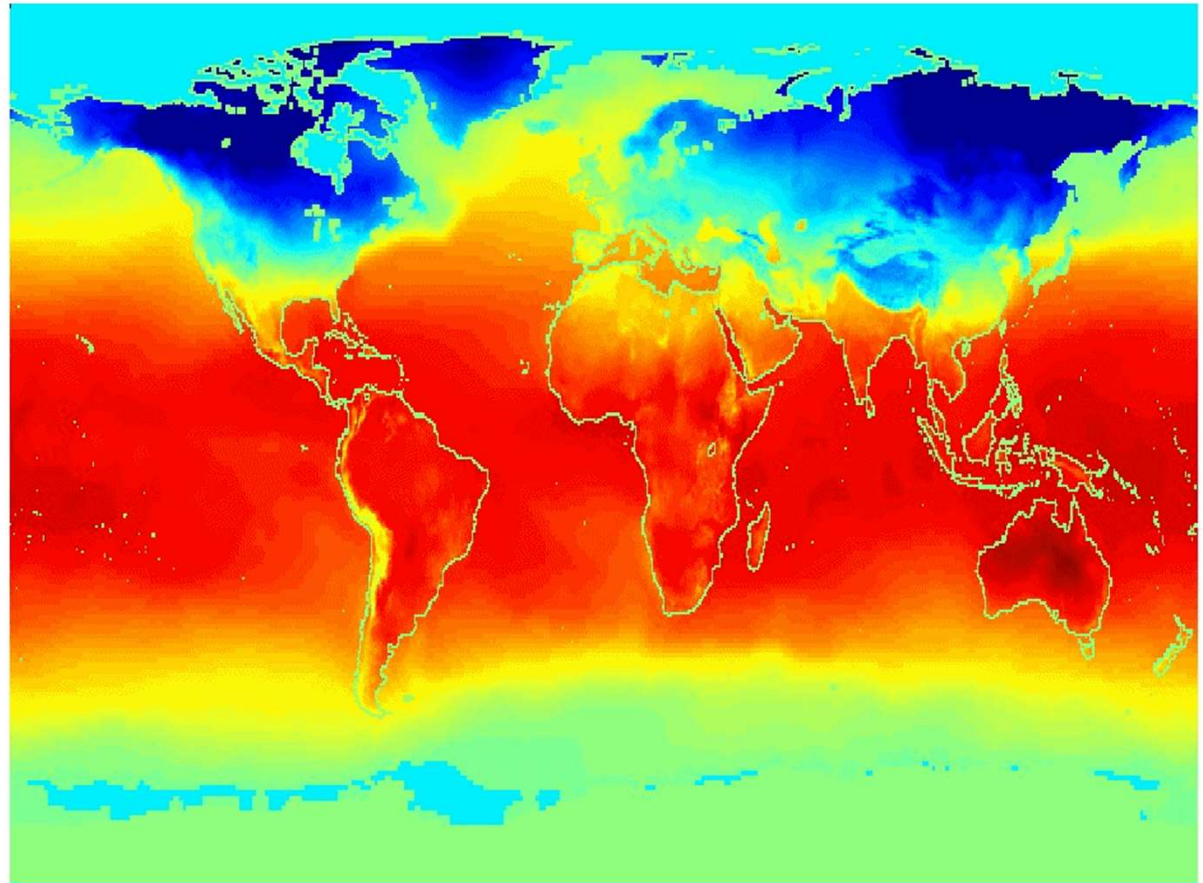
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG

Ordered Data

Spatio-Temporal Data

Average Monthly
Temperature of
land and ocean

Jan



Data Quality

What kinds of data quality problems?

How can we detect problems with the data?

What can we do about these problems?

Examples of data quality problems:

- Noise and outliers

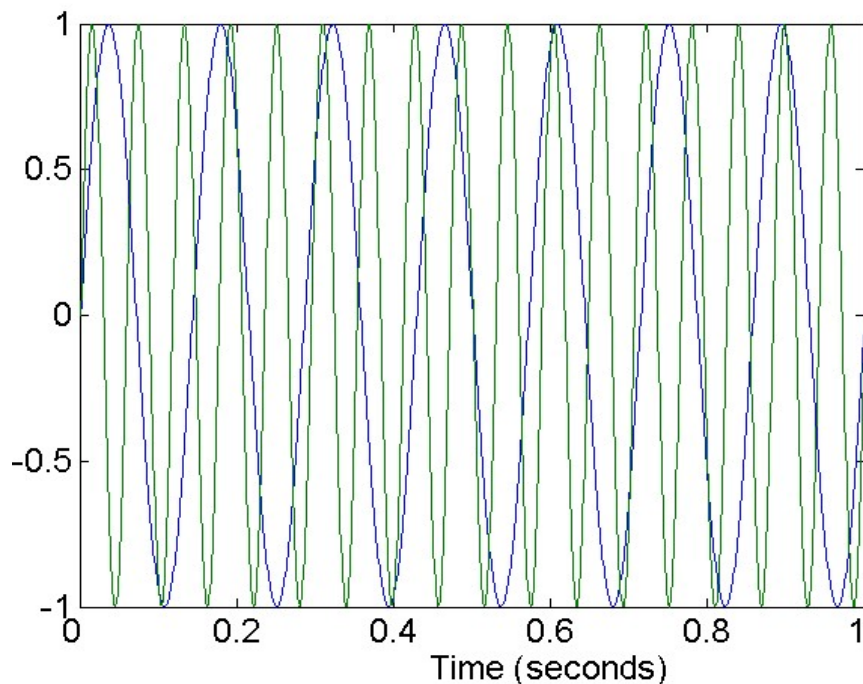
- missing values

- duplicate data

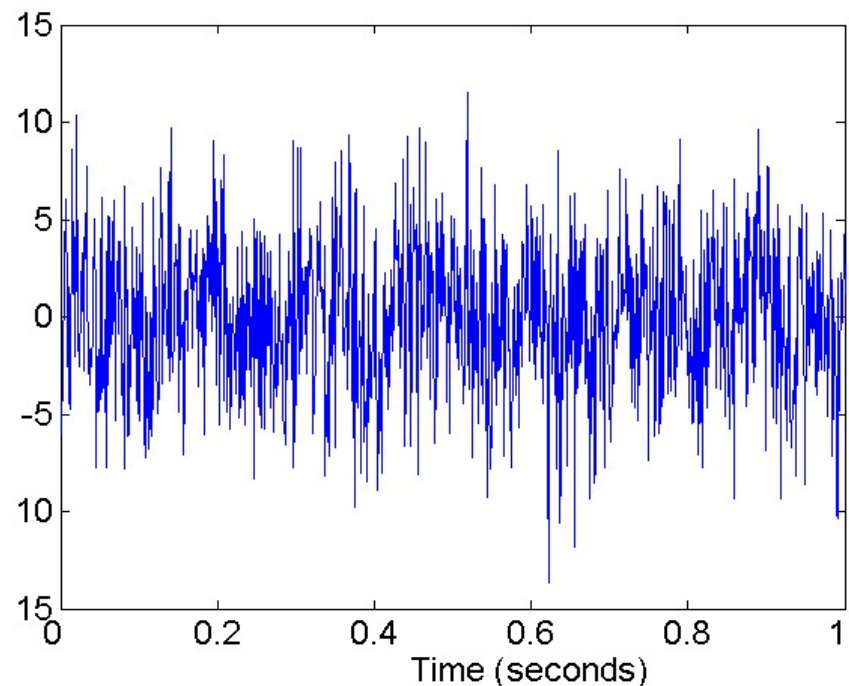
Noise

Noise refers to modification of original values

Examples: distortion of a person's voice when talking on a poor phone and “snow” on television screen



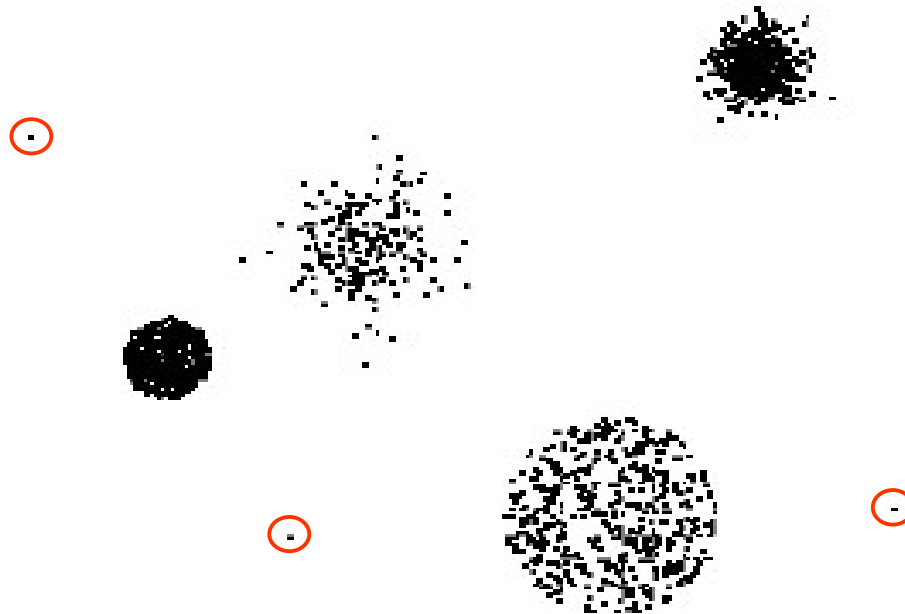
Two Sine Waves



Two Sine Waves + Noise

Outliers

Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set



Missing Values

Reasons for missing values

- Information is not collected
(e.g., people decline to give their age and weight)
- Attributes may not be applicable to all cases
(e.g., annual income is not applicable to children)

Handling missing values

- Eliminate Data Objects
- Estimate Missing Values
- Ignore the Missing Value During Analysis
- Replace with all possible values (weighted by their probabilities)

Duplicate Data

Data set may include data objects that are duplicates, or almost duplicates of one another

Major issue when merging data from heterogeneous sources

Examples:

Same person with multiple email addresses

Data cleaning

Process of dealing with duplicate data issues

Data Preprocessing

- ❖ Aggregation
- ❖ Sampling
- ❖ Dimensionality Reduction
- ❖ Feature subset selection
- ❖ Feature creation
- ❖ Attribute Transformation
- ❖ Discretization and Binarization

Aggregation

Combining two or more attributes (or objects) into a single attribute (or object)

Purpose

Data reduction

Reduce the number of attributes or objects

Change of scale

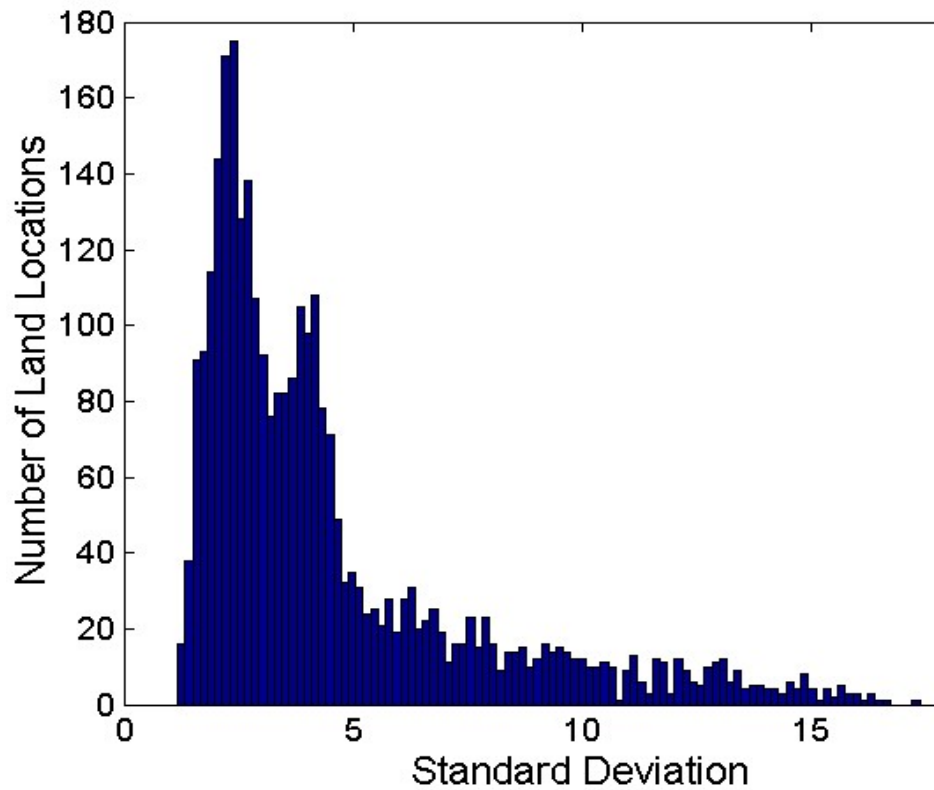
Cities aggregated into regions, states, countries, etc

More “stable” data

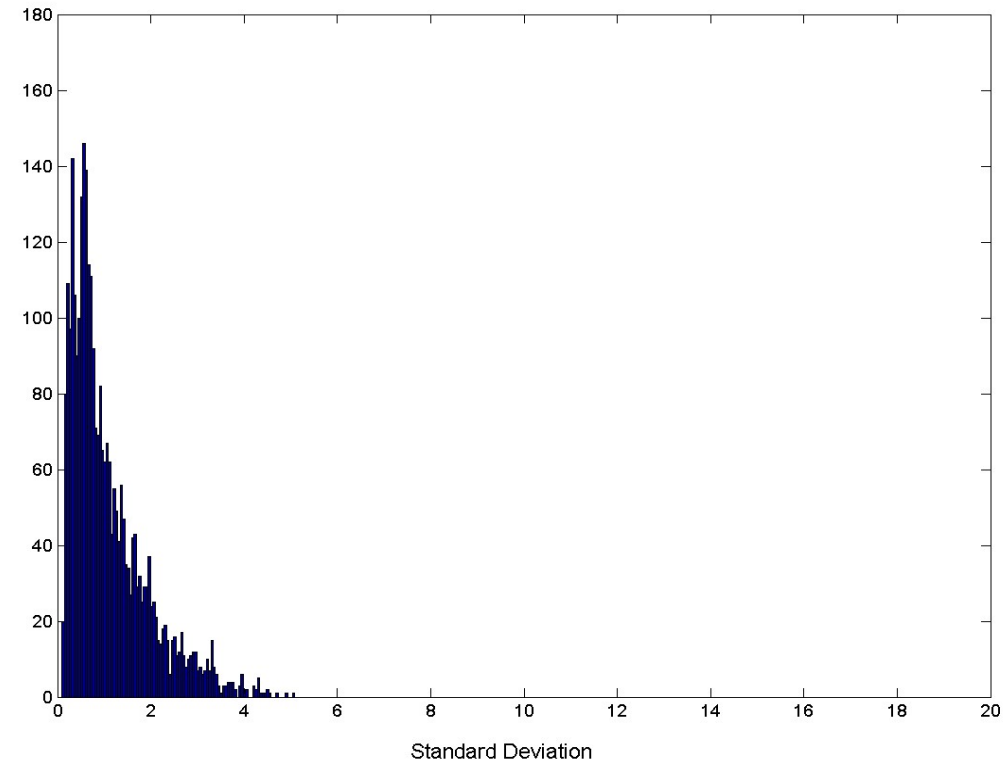
Aggregated data tends to have less variability

Aggregation

Variation of Precipitation in Australia



Standard Deviation of Average
Monthly Precipitation



Standard Deviation of Average
Yearly Precipitation

Sampling

Sampling is the main technique employed for data selection.

It is often used for both the preliminary investigation of the data and the final data analysis.

Statisticians sample because **obtaining** the entire set of data of interest is too expensive or time consuming.

Sampling is used in data mining because **processing** the entire set of data of interest is too expensive or time consuming.

Sampling ...

The key principle for effective sampling is the following:

- using a sample will work almost as well as using the entire data sets, if the sample is representative
- A sample is representative if it has approximately the same property (of interest) as the original set of data

Types of Sampling

Simple Random Sampling

There is an equal probability of selecting any particular item

Sampling without replacement

As each item is selected, it is removed from the population

Sampling with replacement

Objects are not removed from the population as they are selected for the sample.

In sampling with replacement, the same object can be picked up more than once

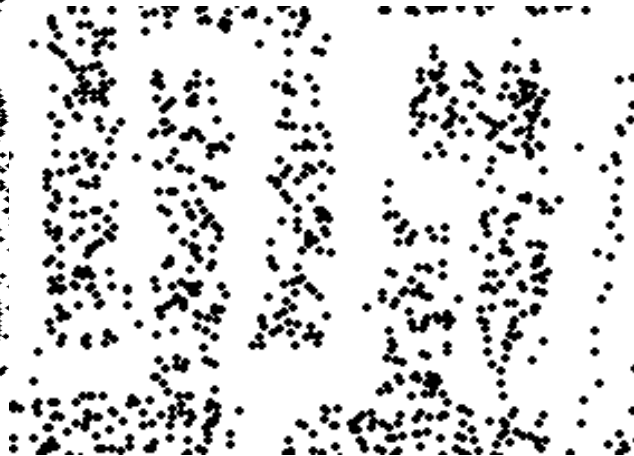
Stratified sampling

Split the data into several partitions; then draw random samples from each partition

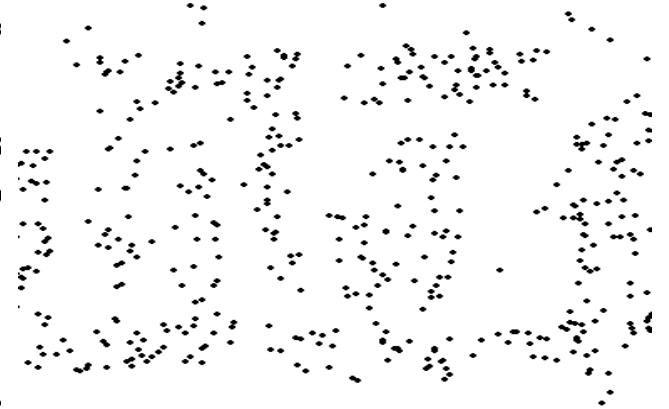
Sample Size



8000 points



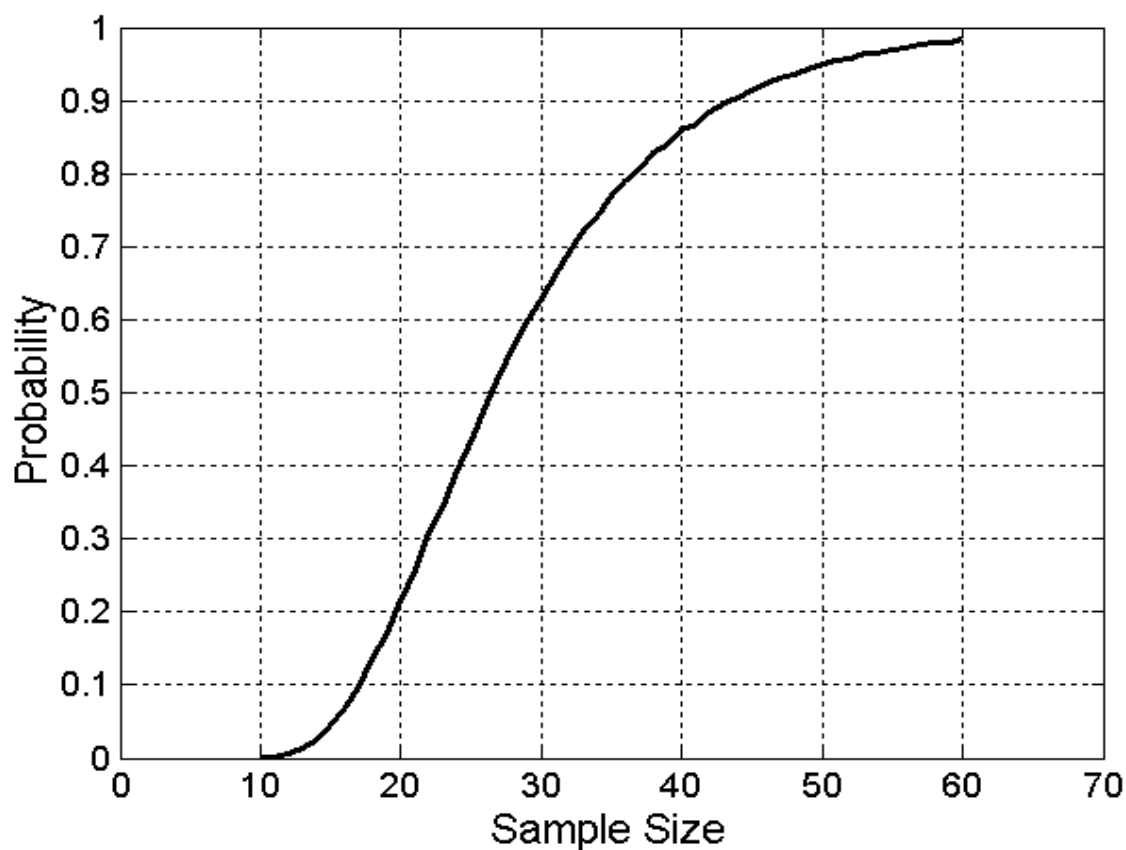
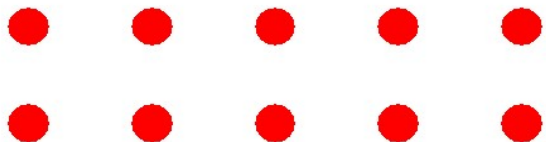
2000 Points



500 Points

Sample Size

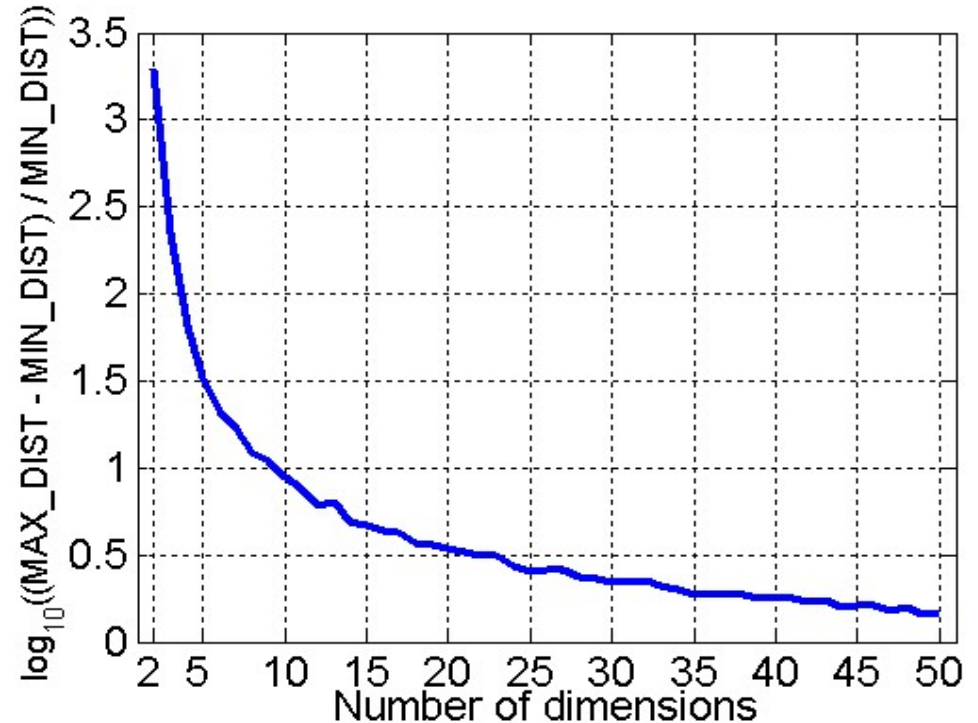
What sample size is necessary to get at least one object from each of 10 groups.



Curse of Dimensionality

When dimensionality increases, data becomes increasingly sparse in the space that it occupies

Definitions of density and distance between points, which is critical for clustering and outlier detection, become less meaningful



- Randomly generate 500 points
- Compute difference between max and min distance between any pair of points

Dimensionality Reduction

Purpose:

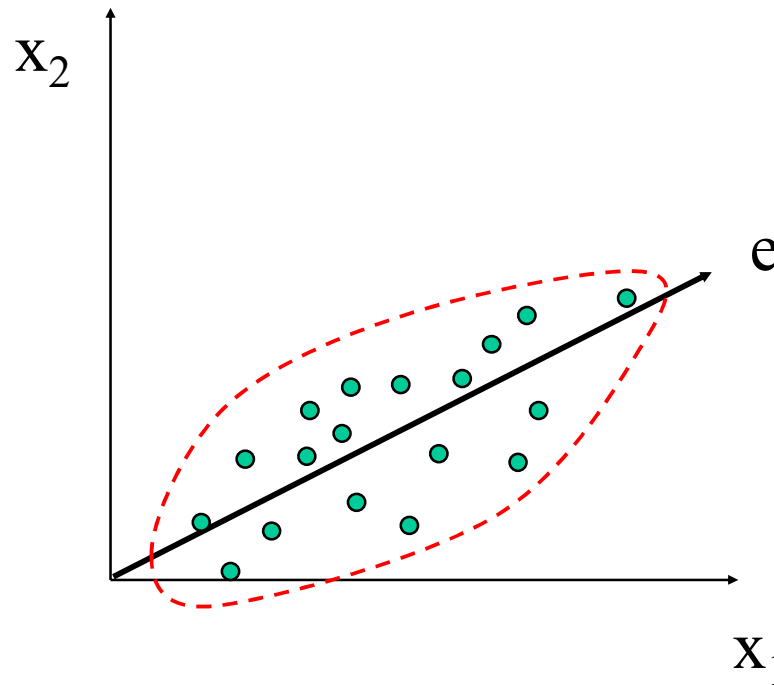
- Avoid curse of dimensionality
- Reduce amount of time and memory required by data mining algorithms
- Allow data to be more easily visualized
- May help to eliminate irrelevant features or reduce noise

Techniques:

- Principal Component Analysis
- Singular Value Decomposition
- Others: supervised and non-linear techniques

Dimensionality Reduction: PCA

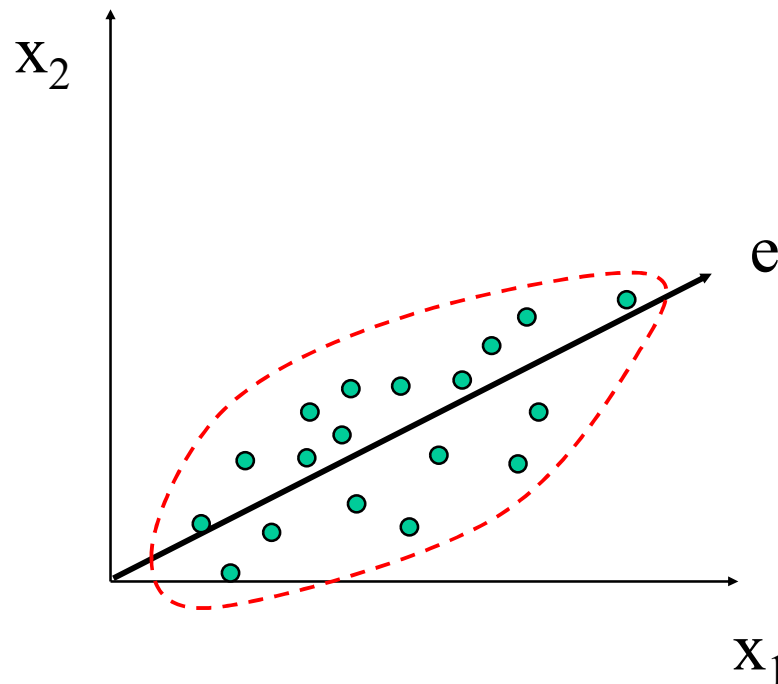
Goal is to find a projection that captures the largest amount of variation in data



Dimensionality Reduction: PCA

Find the eigenvectors of the covariance matrix

The eigenvectors define the new space



Feature Subset Selection

Another way to reduce dimensionality of data

Redundant features

- duplicate much or all of the information contained in one or more other attributes
- Example: purchase price of a product and the amount of sales tax paid

Irrelevant features

- contain no information that is useful for the data mining task at hand
- Example: students' ID is often irrelevant to the task of predicting students' GPA

Feature Subset Selection

Techniques:

- Brute-force approach:

Try all possible feature subsets as input to data mining algorithm

- Embedded approaches:

Feature selection occurs naturally as part of the data mining algorithm

- Filter approaches:

Features are selected before data mining algorithm is run

- Wrapper approaches:

Use the data mining algorithm as a black box to find best subset of attributes

Feature Creation

Create new attributes that can capture the important information in a data set much more efficiently than the original attributes

Three general methodologies:

- Feature Extraction

 - domain-specific

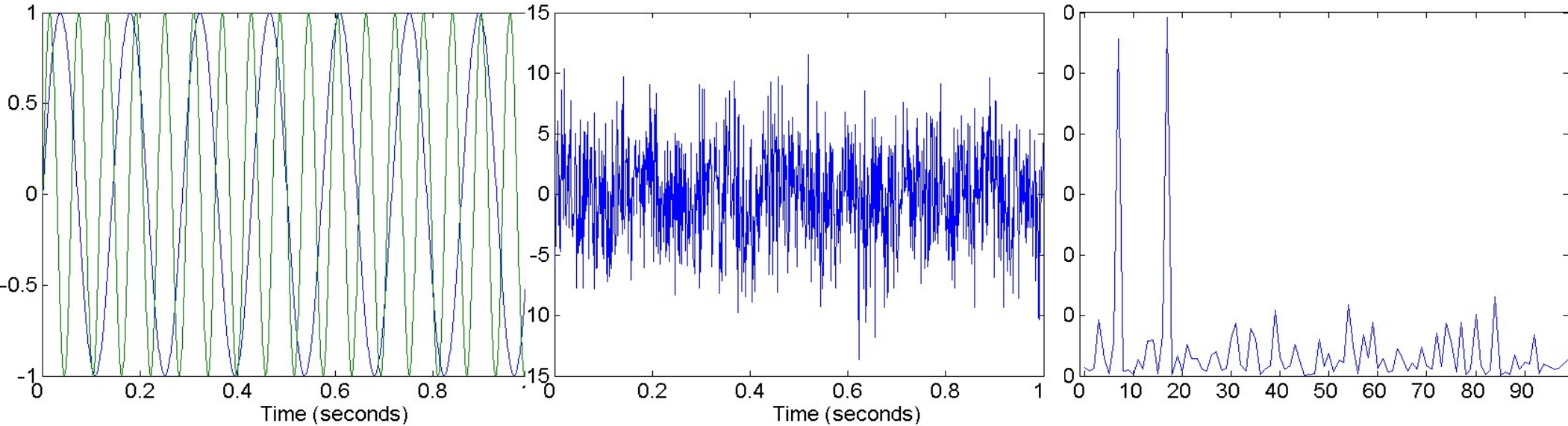
- Mapping Data to New Space

- Feature Construction

 - combining features

Mapping Data to a New Space

- Fourier transform
- Wavelet transform



Two Sine Waves

Two Sine Waves + Noise

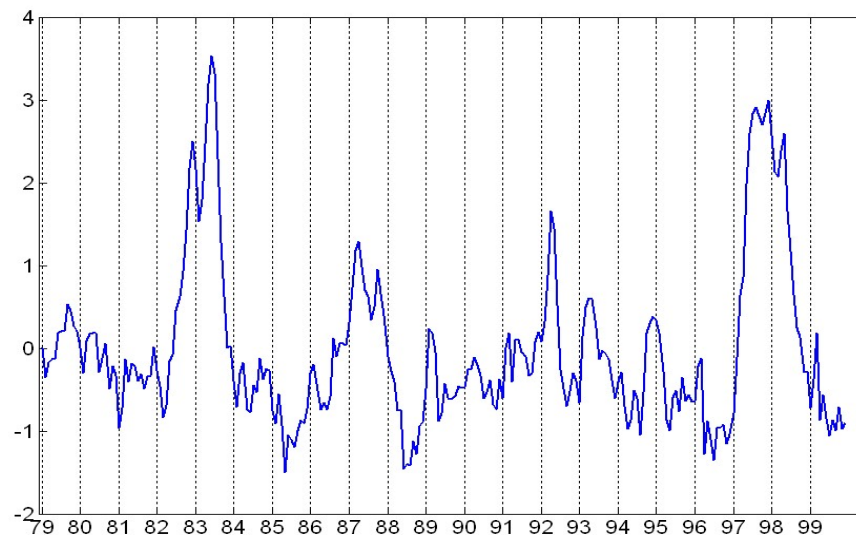
Frequency

Attribute Transformation

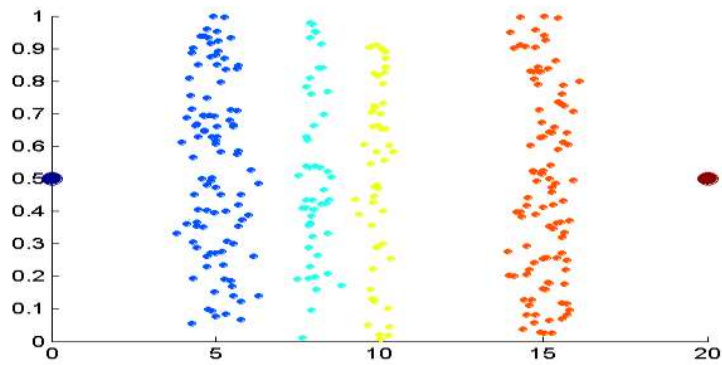
A function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values

Simple functions: x^k , $\log(x)$, e^x , $|x|$

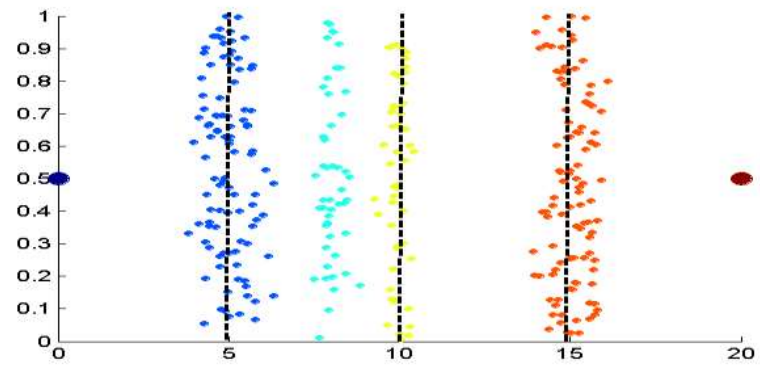
Standardization and Normalization



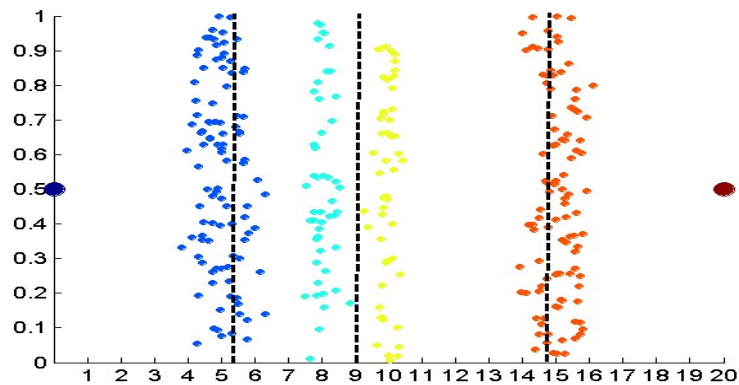
Discretization Without Using Class Labels



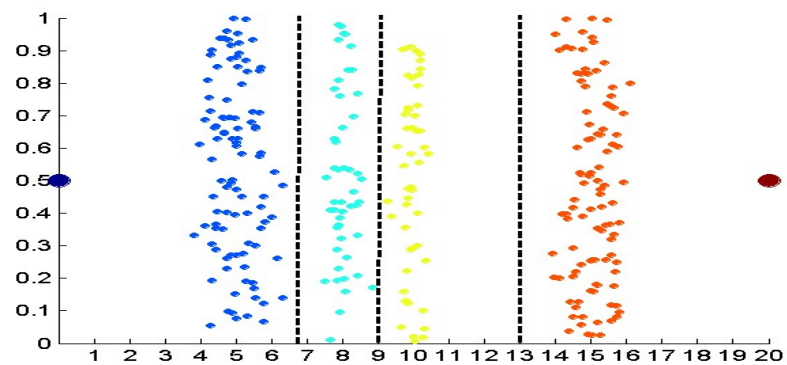
Data



Equal interval width



Equal frequency



K-means

Unsupervised Attribute Filters

- Adding and Removing Attributes
- Changing values
- Conversions
- String conversion
- Multi-Instance data
- Time Series
- Randomizing
- Subsampling
- Normalize

Similarity and Dissimilarity

Similarity

- Numerical measure of how alike two data objects are.
- Is higher when objects are more alike.
- Often falls in the range $[0,1]$

Dissimilarity

- Numerical measure of how different are two data objects
- Lower when objects are more alike
- Minimum dissimilarity is often 0
- Upper limit varies

Proximity refers to a similarity or dissimilarity

Similarity/Dissimilarity for Simple Attributes

p and q are the attribute values for two data objects.

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } p = q \\ 1 & \text{if } p \neq q \end{cases}$	$s = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{if } p \neq q \end{cases}$
Ordinal	$d = \frac{ p-q }{n-1}$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - \frac{ p-q }{n-1}$
Interval or Ratio	$d = p - q $	$s = -d, s = \frac{1}{1+d} \text{ or } s = 1 - \frac{d - \min_d}{\max_d - \min_d}$

Table 5.1. Similarity and dissimilarity for simple attributes

Euclidean Distance

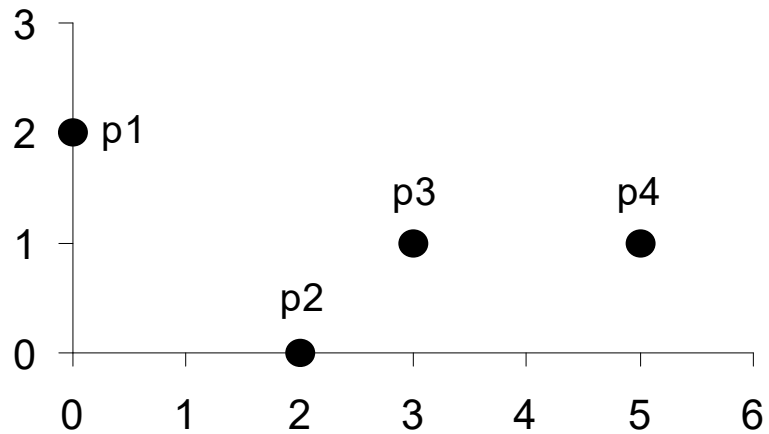
Euclidean Distance

$$\mathbf{dist} = \sqrt{\sum_{k=1}^n (\mathbf{p}_k - \mathbf{q}_k)^2}$$

Where n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k^{th} attributes (components) or data objects p and q .

Standardization is necessary, if scales differ.

Euclidean Distance



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Distance Matrix

Minkowski Distance

Minkowski Distance is a generalization of Euclidean Distance

$$\mathbf{dist} = \left(\sum_{k=1}^n | \mathbf{p}_k - \mathbf{q}_k |^r \right)^{\frac{1}{r}}$$

Where r is a parameter, n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k -th attributes (components) or data objects p and q .

Minkowski Distance: Examples

$r = 1$. City block (Manhattan, taxicab, L_1 norm) distance.

A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors

$r = 2$. Euclidean distance

$r \rightarrow \infty$. “supremum” ($L_{\max}$ norm, L_{∞} norm) distance.

This is the maximum difference between any component of the vectors

Do not confuse r with n , i.e., all these distances are defined for all numbers of dimensions.

Minkowski Distance

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_∞	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Distance Matrix

Common Properties of a Distance

Distances, such as the Euclidean distance, have some well known properties.

1. $d(p, q) \geq 0$ for all p and q and $d(p, q) = 0$ only if $p = q$. (Positive definiteness)
2. $d(p, q) = d(q, p)$ for all p and q . (Symmetry)
3. $d(p, r) \leq d(p, q) + d(q, r)$ for all points p, q , and r . (Triangle Inequality)

where $d(p, q)$ is the distance (dissimilarity) between points (data objects), p and q .

A distance that satisfies these properties is a metric

Common Properties of a Similarity

Similarities, also have some well known properties.

1. $s(p, q) = 1$ (or maximum similarity) only if $p = q$.
2. $s(p, q) = s(q, p)$ for all p and q . (Symmetry)

where $s(p, q)$ is the similarity between points (data objects), p and q .

Similarity Between Binary Vectors

Common situation is that objects, p and q , have only binary attributes

Compute similarities using the following quantities

M_{01} = the number of attributes where p was 0 and q was 1

M_{10} = the number of attributes where p was 1 and q was 0

M_{00} = the number of attributes where p was 0 and q was 0

M_{11} = the number of attributes where p was 1 and q was 1

Simple Matching and Jaccard Coefficients

SMC = number of matches / number of attributes

$$= (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00})$$

J = number of M11 matches / number of not-both-zero attributes values

$$= (M_{11}) / (M_{01} + M_{10} + M_{11})$$

SMC versus Jaccard: Example

$p = 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0$

$q = 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 1$

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)

$M_{10} = 1$ (the number of attributes where p was 1 and q was 0)

$M_{00} = 7$ (the number of attributes where p was 0 and q was 0)

$M_{11} = 0$ (the number of attributes where p was 1 and q was 1)

$$SMC = (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) = (0+7) / (2+1+0+7) = 0.7$$

$$J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = 0 / (2 + 1 + 0) = 0$$

Cosine Similarity

If d_1 and d_2 are two document vectors, then

$$\cos(d_1, d_2) = (d_1 \bullet d_2) / ||d_1|| ||d_2|| ,$$

where $\bullet$ indicates vector dot product and $||d||$ is the length of vector d .

Example:

$$d_1 = \mathbf{3\ 2\ 0\ 5\ 0\ 0\ 0\ 2\ 0\ 0}$$

$$d_2 = \mathbf{1\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 2}$$

$$d_1 \bullet d_2 = 3*1 + 2*0 + 0*0 + 5*0 + 0*0 + 0*0 + 0*0 + 2*1 + 0*0 + 0*2 = 5$$

$$||d_1|| = (3*3 + 2*2 + 0*0 + 5*5 + 0*0 + 0*0 + 0*0 + 2*2 + 0*0 + 0*0)^{0.5} = (42)^{0.5} \\ = 6.481$$

$$||d_2|| = (1*1 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 1*1 + 0*0 + 2*2)^{0.5} = (6)^{0.5} \\ = 2.245$$

$$\cos(d_1, d_2) = .3150$$

Correlation

Correlation measures the linear relationship between objects

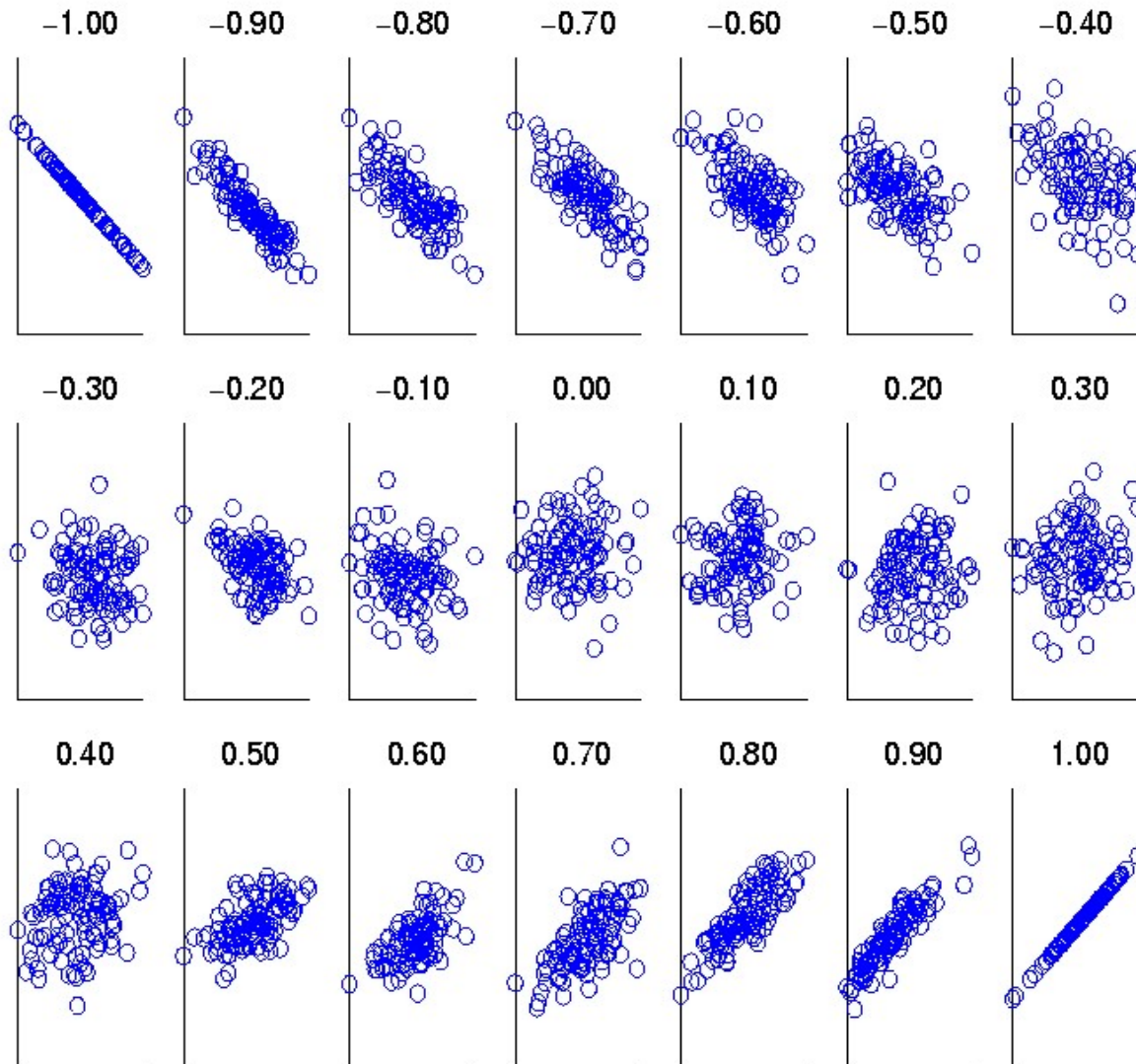
To compute correlation, we standardize data objects, p and q , and then take their dot product

$$p'_k = (p_k - \text{mean}(p)) / \text{std}(p)$$

$$q'_k = (q_k - \text{mean}(q)) / \text{std}(q)$$

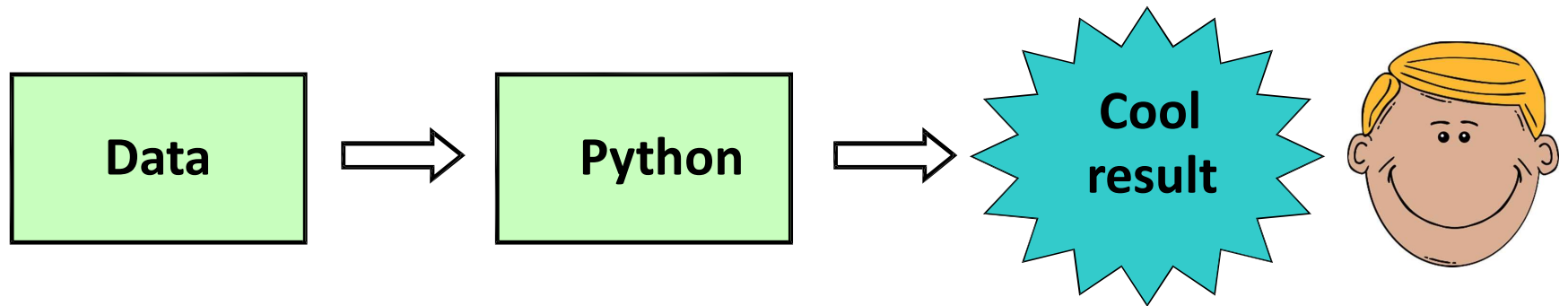
$$\text{correlation}(p, q) = p' \bullet q'$$

Visually Evaluating Correlation

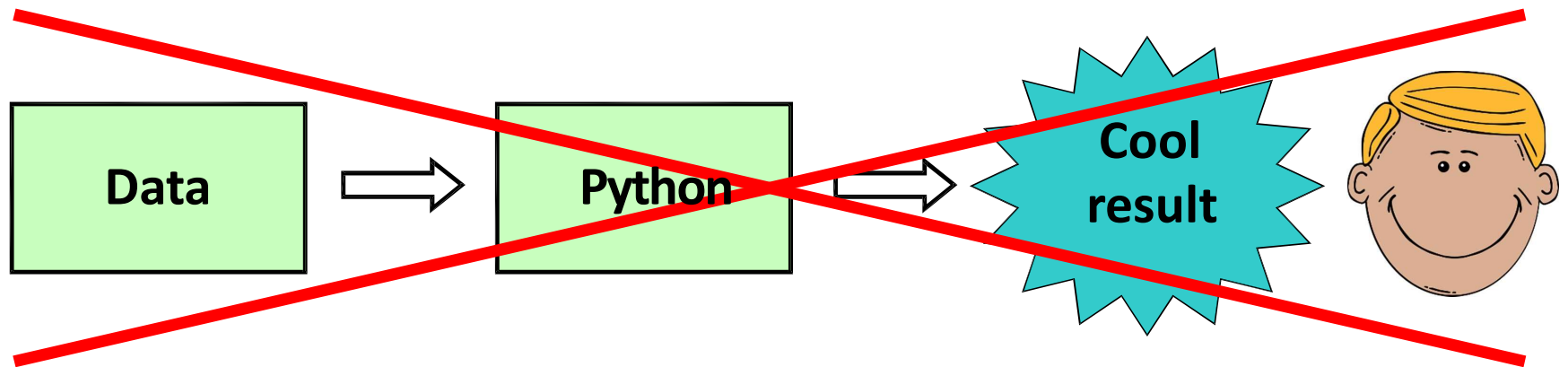


Scatter plots showing the similarity from -1 to 1.

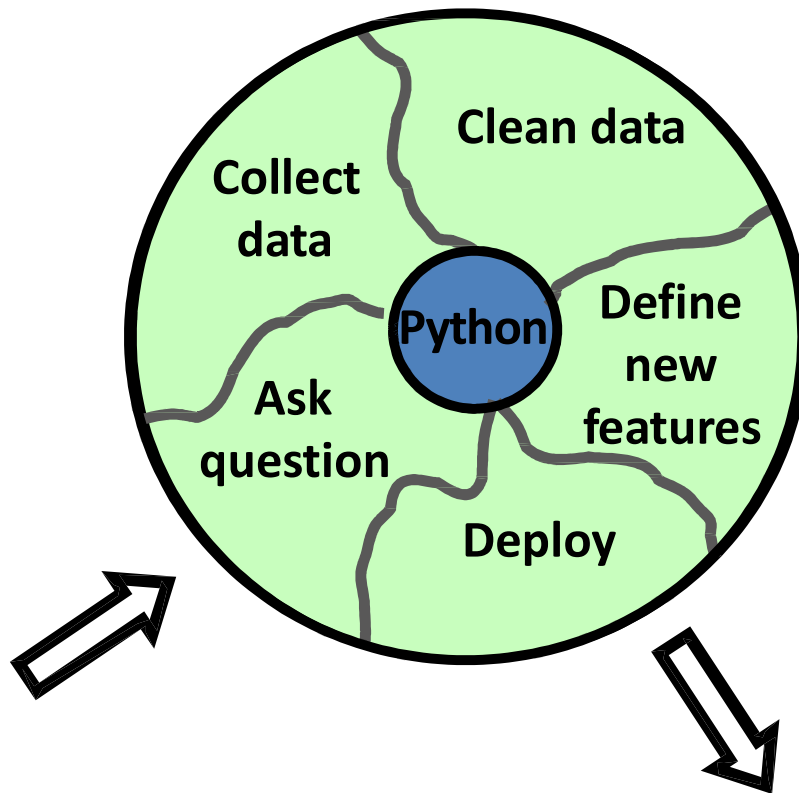
The data mining process



The data mining process



The data mining process



The data mining process

- ❖ Ask a question
 - *what do you want to know?*
 - *“tell me something cool about the data” is not enough!*
- ❖ Gather data
 - *there’s soooo much around ...*
 - *... **but** ... we need (expert?) classifications more data beats*
 - *a clever algorithm*
- ❖ Clean the data
 - *real data is very mucky*
- ❖ Define new features
 - *feature engineering—the key to data mining*
- ❖ Deploy the result
 - *technical implementation*
 - *convince your boss!*

The data mining process

(Selected) filters for feature engineering

- ❖ AddExpression (MathExpression)
 - Apply a math expression to existing attributes to create new one (or modify existing one)*
- ❖ Center (Normalize) (Standardize)
 - Transform numeric attributes to have zero mean (or into a given numeric range) (or to have zero mean and unit variance)*
- ❖ Discretize (also supervised discretization)
 - Discretize numeric attributes to have nominal values*
- ❖ PrincipalComponents
 - Perform a principal components analysis/transformation of the data*
- ❖ RemoveUseless
 - Remove attributes that do not vary at all, or vary too much*
- ❖ TimeSeriesDelta, TimeSeriesTranslate
 - Replace attribute values with successive differences between this instance and the next*

Pitfalls and pratfalls

Pitfall: A hidden or unsuspected danger or difficulty

Pratfall: A stupid and humiliating action

Pitfalls and pratfalls

Be skeptical

- ❖ In data mining, it's very easy to cheat
 - *whether consciously or unconsciously*
- ❖ For reliable tests, use a completely fresh sample of data that has never been seen before

Overfitting has many faces

- ❖ Don't test on the training set (of course!)
- ❖ Data that has been used for development (in any way) is tainted
- ❖ Leave some evaluation data aside for the *very end*

Pitfalls and pratfalls

Missing values

“Missing” means what ...

- ❖ Unknown?
- ❖ Unrecorded?
- ❖ Irrelevant?

Should you: 1. Omit instances where the attribute value is missing?
or 2. Treat “missing” as a separate possible value?

Is there significance in the fact that a value is missing?

Most learning algorithms deal with missing values

– but they may make different assumptions about them

Pitfalls and pratfalls

No free lunch



- ❖ 2-class problem with 100 binary attributes
- ❖ Say you know a million instances, and their classes (training set)
- ❖ You don't know the classes of $2^{100} - 10^6$ examples!
(that's 99.9999...% of the data set)
- ❖ How could you possibly figure them out?

In order to generalize, every learner must embody some knowledge or assumptions beyond the data it's given

A learning algorithm implicitly provides a set of assumptions
There can be no “universal” best algorithm (no free lunch)

Data mining is an experimental science

Pitfalls and pratfalls

- ❖ Be skeptical
- ❖ Overfitting has many faces
- ❖ Missing values – different assumptions
- ❖ No “universal” best learning algorithm
- ❖ Data mining is an experimental science
- ❖ It's very easy to be misled

Data mining and ethics

Information privacy laws (in Europe, but not US)

- ❖ A purpose must be stated for any personal information collected
- ❖ Such information must not be disclosed to others without consent
- ❖ Records kept on individuals must be accurate and up to date
- ❖ To ensure accuracy, individuals should be able to review data about themselves
- ❖ Data must be deleted when it is no longer needed for the stated purpose
- ❖ Personal information must not be transmitted to locations where equivalent data protection cannot be assured
- ❖ Some data is too sensitive to be collected, except in extreme circumstances (e.g., sexual orientation, religion)

Data mining and ethics

Anonymization is harder than you think

When Massachusetts released medical records summarizing every state employee's hospital record in the mid-1990s, the governor gave a public assurance that it had been anonymized by removing all identifying information such as name, address, and social security number. He was surprised to receive his own health records (which included diagnoses and prescriptions) in the mail.

Data mining and ethics

Reidentification techniques. Using publicly available records:

- ❖ 50% of Americans can be identified from city, birth date, and sex
- ❖ 85% can be identified if you include the 5-digit zip code as well

Netflix movie database: 100 million records of movie ratings (1–5)

- ❖ Can identify 99% of people in the database if you know their ratings for 6 movies and approximately when they saw the movies ($\pm$ one week)
- ❖ Can identify 70% if you know their ratings for 2 movies and roughly when they saw them

Data mining and ethics

The purpose of data mining is to discriminate ...

- ❖ who gets the loan
- ❖ who gets the special offer

Certain kinds of discrimination are unethical, and illegal

- ❖ racial, sexual, religious, ...

Data mining and ethics

The purpose of data mining is to discriminate ...

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Certain kinds of discrimination are unethical, and illegal

- ❖ racial, sexual, religious, ...

But it depends on the context

- ❖ sexual discrimination is usually illegal
- ❖ ... except for doctors, who are *expected* to take gender into account

... and information that appears innocuous may not be

- ❖ ZIP code correlates with race
- ❖ membership of certain organizations correlates with gender

Data mining and ethics

Correlation does not imply causation

As icecream sales increase, so does the rate of drownings.

Therefore icecream consumption causes drowning???

Data mining reveals correlation, not causation

but really, we want to predict the effects of our actions

Summary

- ❖ **There's no magic in data mining**
 - *Instead, a huge array of alternative techniques*
- ❖ **There's no single universal “best method”**
 - *It's an experimental science!*
 - *What works best on your problem?*
- ❖ **Weka makes it easy**
 - *... maybe too easy?*
- ❖ **There are many pitfalls**
 - *You need to understand what you're doing!*
- ❖ **Focus on evaluation ... and significance**
 - *Different algorithms differ in performance – but is it significant?*